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CSE 242 Assignment 3.

Q 1 Hard-margin linear SVM

a) positive class (points 1, 2, 3) sit in the upper-right
negative class (points 4, 5, 6) are in the lower left of the plot.

The support vectors are the points of each class that are closest to the other class.

∴ a line from upper left to lower right diagonal will be a good separation line.

On the positive side: point 1 (1, 1) is the closest to the negative side.

on the negative side: points 6 (1, 0) & point 5 (0, 1) are the closest to the positive side.

∴ the widest separation line runs between these 2 lines with the decision boundary.

1b. Derivation.

→

For a hard-margin SVM the support vectors satisfy $w \cdot x + b = +1$ (positive class) and $w \cdot x + b = -1$ (negative) class

Substitute the three support vectors:

point 1, (1,1): $w_1 + w_2 + b = 1$ (1)

point 5, (1,0): $w_1 + b = -1$ (2)

point 6, (0,1): $w_2 + b = -1$ (3)

equation (2) - (3) gives $w_1 = w_2 = a$

Then (1) is $2a + b = 1$ & (2) is $a + b = 1$

subtracting gives $a = 2$ & $b = -3$

$$\boxed{w = 2, 2}$$

$$\boxed{b = -3}$$

boundary: $2x_1 + 2x_2 - 3 = 0$ (i.e. $x_1 + x_2 = 1.5$)

Geometric Margin: the distance from the boundary to a support vectors is $\frac{1}{\|w\|}$ and

$$\|w\| = \sqrt{2^2 + 2^2} = 2\sqrt{2}$$

$$\therefore \boxed{\gamma = \frac{1}{\|w\|} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4} = 0.0356}$$

the full width between the 2 lines is

$$\frac{2}{\|w\|} = \boxed{0.7071}$$

Q2.

a)

Tie rule: the comparison is strictly $<$ so a point equidistant from several centroids stays with the first and lowest centroid.

b) Empty cluster policy: a cluster with no points keeps its previous centroid.

Initialize samples rows with replacement so duplicate initial centroids are possible.

Q2.

b)

$K=2$ isolates the top blob and lumps the other three together.

$K=3$ merges the 2 adjacent groups in the middle blobs

$K=4$ is the most accurate as it separates the classes accurately.

at $K=5$ the a single blob starts to split into smaller groups

Similarly higher K values such as K=6 & K=10 split the blobs into segments.

Therefore, a small K merges real clusters, K near to the true number of clusters performs best.

K higher value fragments the blobs into smaller groups.

Q3.

a & b.

repair_labels(k=5) fits Nearest Neighbours on the training features, find each point's 5 nearest neighbours.

All votes read from the same original noisy array.

The method only uses training features & noisy labels.

A fresh SVM with same params is trained as baseline.

the results:

Clean label reference accuracy : 0.8670

Baseline (noisy labels) mean : 0.8280 (std 0.0179)

K-NN repaired labels mean : 0.8644 (std 0.100)

difference (KNN - Baseline) : +0.0364

Did repair help? Yes the repair of SVM beat the baseline always. By 3.6 points.

Why did it help?

- The flips are random & independent so a flipped point is usually surrounded by correctly labeled neighbors of its own class.

Repair also helps in solving class imbalance problems. the asymtric noise cruely.

Here it solves the problem of overlapping classes.